

Learner-Risk Analytics for Workforce-Oriented Postsecondary Pathways: A Pre-Pilot Evaluation Framework

Abstract

Institutions can now generate learner-risk scores faster than they can decide whether those scores are calibrated, interpretable, equitable, staffable, and appropriate for learner-facing use. This study presents WALA-CC, a pre-pilot evaluation framework for learner-risk analytics in workforce-oriented postsecondary pathways. The framework combines six open learner benchmark datasets totaling 38,817 records, public two-year institutional indicators, and occupational skill descriptors while keeping learner, institution, program, and occupation units analytically separate. The reproducible pipeline compares majority and rule baselines with elastic-net logistic regression, random forest, extra trees, histogram gradient boosting, calibrated gradient boosting, XGBoost, and LightGBM using repeated stratified 5x5 cross-validation, bootstrap intervals, ROC curves, calibration diagnostics, feature-block ablation, subgroup checks, and learner-profile analysis. Across 54 learner-model combinations, leading held-out macro-F1 ranged from 0.773 to 0.949. UCI outcome-coding sensitivity showed that excluding still-enrolled cases raised the best UCI macro-F1 from 0.826 to 0.902, reinforcing the need to inspect label assumptions. Capacity-aware scenarios translated calibrated probabilities into learners flagged per 100, advisor hours, projected additional progression under explicit support assumptions, and subgroup flag-gap diagnostics. The contribution is not a deployment claim, but an evidence-rich workflow for deciding what institutions should inspect before piloting learner-risk analytics for adaptive support.

Keywords: learning analytics, learner-risk modeling, adaptive support, workforce education, community colleges, AI governance

1 Introduction

Artificial intelligence has intensified a long-standing challenge in educational technology: institutions can generate learner-risk scores more easily than they can decide how those scores should be interpreted, governed, and acted upon. This tension is consequential for workforce-oriented postsecondary pathways. Community colleges are a motivating case because they combine open access, transfer programs, career and technical education, noncredit training, and regional workforce missions while serving large

numbers of part-time, working, first-generation, and financially constrained learners (American Association of Community Colleges, 2025; Bailey, Jaggars, & Jenkins, 2015). Current completion evidence also shows why timely support matters. National Student Clearinghouse reporting indicates that public two-year completion remains lower than four-year-sector completion even when completions after transfer are counted (Lee & Shapiro, 2023). Delayed support can therefore become an institutional design problem rather than only an individual learner problem.

Learning analytics and educational data mining have shown that digital activity, assessment, registration, finance, and background variables can be associated with course progression and study success (Bulut, Shin, Yildirim-Erbasli, & Gorgun, 2023; Ifenthaler & Yau, 2020; Park, Yu, Tang, Liu, & Liu, 2023; Siemens, 2013). Yet the AI-era question is no longer only whether a model can classify learners. Before a college pilots learner-risk analytics, it should know whether the probabilities are calibrated, whether early signals arrive soon enough for action, whether explanations are educationally meaningful, whether subgroup errors are visible, whether alert volumes can be staffed, and whether learners retain agency when analytics shape outreach. Without these checks, high discrimination can hide poor calibration, a plausible threshold can exceed advising capacity, and a support narrative can rest on untested assumptions.

This article introduces WALA-CC, a multi-source pre-pilot evaluation framework for learner-risk analytics in workforce-oriented postsecondary pathways. Community and technical colleges provide the motivating institutional context, but the learner-level models are not estimated from U.S. community-college administrative records. Instead, the study uses open learner benchmarks to pressure-test risk modeling and diagnostics, public U.S. two-year institution indicators to describe institutional context, and CIP-SOC/O*NET records to make workforce-context assumptions auditable. These sources are linked by analytic role rather than by row-level merging. This boundary of inference is central: learner records, institution aggregates, program-occupation crosswalks, and occupational skill descriptors refer to different units, and a forced merge would imply information that the public data do not contain.

The study makes three contributions to *Education and Information Technologies* scholarship. First, it provides an education-facing evaluation workflow that joins discrimination, calibration, early-action validity, explainability, subgroup diagnostics, learner profiles, and capacity analysis rather than presenting a model leaderboard alone. Second, it translates calibrated learner-risk probabilities into support queues, advisor-hour demand, and scenario-projected progression under explicit, non-causal support assumptions. Third, it downgrades workforce evidence from an outcome claim to workforce-context auditability: public program-occupation and skill data are used to document pathway context, not to predict individual employment or readiness.

The study is guided by four research questions:

1. What level of learner-risk prediction, calibration, and early-action validity can open benchmark learner records support?
2. Which feature families provide action-relevant signal across academic progress, engagement, background, enrollment, and finance variables?
3. What subgroup, learner-profile, and workforce-context diagnostics should institutions inspect before considering an adaptive-support pilot?

4. Under explicit support assumptions, what coverage, projected progression, advisor-hour, and subgroup flag-gap trade-offs emerge from capacity-aware scenarios?

Together, these questions position WALA-CC as a pre-pilot evidence base for institutional researchers, advising leaders, pathway directors, and governance committees. The framework treats learner-risk scores as prompts for supportive inquiry, not as grounds for restricting access, punishing learners, or denying opportunities.

2 Theoretical and Empirical Motivation

2.1 From prediction to pre-pilot readiness

Learning analytics is commonly defined around measuring, collecting, analyzing, and reporting learner data to understand and improve learning environments (Siemens, 2013). Recent work has shifted attention from predictive performance alone toward validation, human-centered use, ethical review, and institutional accountability (Motz & Makany, 2023; Soffer, Kahan, & Nachmias, 2024). This shift matters because colleges often encounter analytics through dashboards, vendor tools, or local predictive models before they have decided how alerts will be interpreted. The gap motivating WALA-CC is therefore not simply a need for stronger classifiers; it is the absence of a compact workflow for judging whether learner-risk analytics are ready for a local pilot.

Figure 1 summarizes the problem orientation used in this study. Learner-risk analytics for workforce-oriented pathways can fail when learner signals are fragmented, probabilities are poorly calibrated, advisor capacity is not modeled, workforce context is treated as a slogan rather than evidence, governance is added late, or subgroup threshold effects are ignored. These risks justify the framework’s emphasis on calibration, early-action timing, explanation, subgroup behavior, capacity, and workforce-context auditability.

2.2 Calibration, explanation, and threshold governance

Many learner-risk studies emphasize discrimination metrics such as accuracy or area under the ROC curve. These metrics are useful, but they are incomplete for adaptive support. If probabilities trigger advising, tutoring, financial-aid outreach, or program review, they should correspond to observed frequencies (Guo, Pleiss, Sun, & Weinberger, 2017; Niculescu-Mizil & Caruana, 2005). A model can rank learners well and still overstate or understate risk in ways that distort support queues. The practical gap is that institutional users need probability-quality and threshold diagnostics in the same report as discrimination metrics.

Explainability creates a second gap. Tree ensembles such as random forests, XGBoost, and LightGBM can perform well on structured educational records, but they need interpretation tools that help educators understand which feature families contribute to risk estimates (Chen & Guestrin, 2016; Ke et al., 2017; Lundberg & Lee, 2017). WALA-CC therefore reports feature-level importance and grouped feature-family contribution. The grouped view is included because an advising lead is more likely to act on a pattern such as first-semester academic progress, financial status, or early virtual learning environment activity than on a single encoded variable.

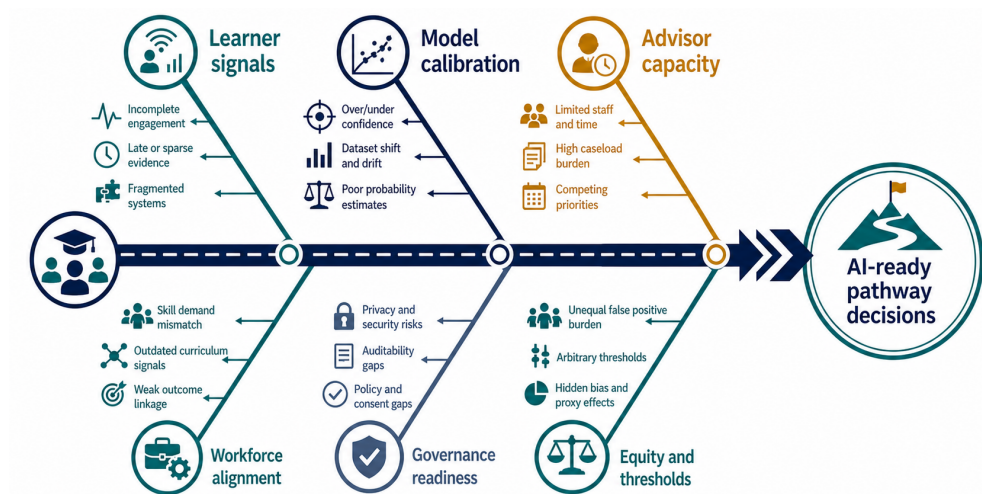


Fig. 1 Problem map for learner-risk analytics in workforce-oriented pathways. The map highlights why learner signals, model calibration, advisor capacity, workforce-context auditability, governance readiness, and equity-sensitive thresholds must be evaluated together before institutional piloting

2.3 Community colleges, support intensity, and learner agency

Community colleges differ from selective four-year contexts in mission, student composition, and operational constraints. They combine open access with transfer pathways, certificate programs, career and technical education, developmental coursework, and regional workforce partnerships ([American Association of Community Colleges, 2025](#); [Bailey et al., 2015](#)). Evidence from CUNY ASAP shows that intensive, multi-year wraparound support can substantially raise completion in a randomized setting, but that intervention is far more comprehensive than a light-touch analytics alert ([Gupta, 2017](#); [Weiss, Ratledge, Sommo, & Gupta, 2019](#)). The gap for learner-risk analytics is that most pre-pilot reports do not translate risk thresholds into staff time, support intensity, or plausible scenario assumptions.

Learner agency is equally important. Responsible analytics should not treat risk estimates as conclusions about learners. Scores should prompt supportive inquiry, learner-facing explanation, data correction, contestability, and escalation pathways when analytics influence outreach ([Kitto & Knight, 2020](#); [Slade & Prinsloo, 2013](#)). This gap justifies a non-punitive design principle in WALA-CC: alerts should not be used to restrict access, deny course choices, punish learners, or replace advisor judgment.

2.4 Workforce-context auditability

Workforce pathway claims require careful unit separation. The NCES CIP-SOC cross-walk links instructional programs to occupations, and O*NET describes occupational skill importance. These sources can help a college document whether a pathway-facing analytics pilot is being discussed in relation to transparent program and occupation evidence. They do not observe individual job placement, wages, licensing, or workforce

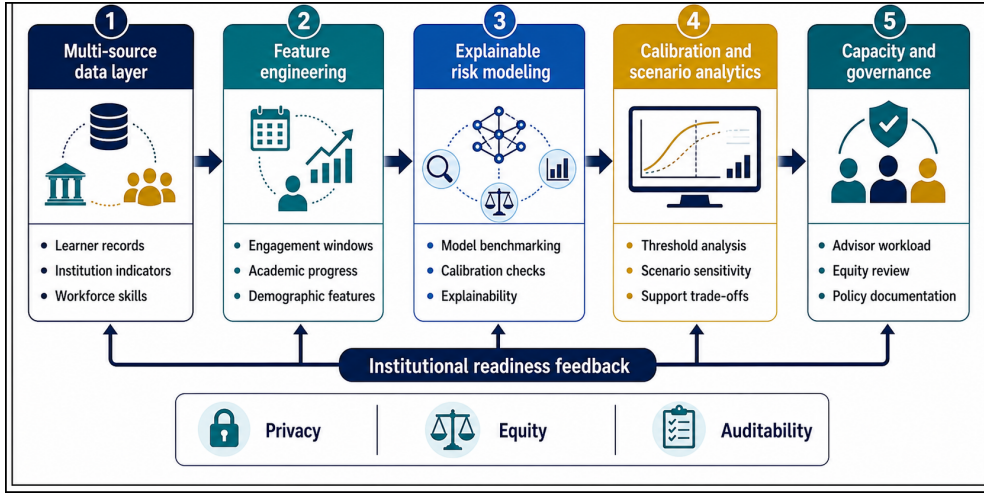


Fig. 2 Architecture of the WALA-CC pre-pilot evaluation framework. Open benchmark learner records support explainable risk modeling, public institutional and workforce indicators provide context, and calibrated probabilities are translated into threshold and capacity scenarios for governance review

readiness. The gap is therefore not a missing learner-to-job prediction model; it is a need for workforce-context auditability that lets institutions document the pathway assumptions surrounding an analytics pilot without creating false individual-level labor-market claims.

3 The WALA-CC Framework

WALA-CC is organized as a layered pre-pilot evaluation architecture for learner-risk analytics in workforce-oriented postsecondary pathways, anchored in the operational realities of community and technical colleges. Figure 2 shows the architecture. The framework begins with separate evidence streams, converts learner records into calibrated progression-risk estimates, adds explainability and subgroup diagnostics, and then translates risk estimates into capacity-aware scenarios that can be reviewed by human decision makers.

The first layer is the multi-source data layer. It treats learner records, institution indicators, and workforce indicators as distinct forms of evidence. Learner records support prediction and subgroup diagnostics. Institution records describe the public two-year college setting. Workforce records describe occupational skill families connected to program areas. The framework avoids row-level merging across these units because such a merge would suggest that institution and occupational attributes were observed for individual learners.

The second layer is feature engineering. For learner datasets, variables are grouped into interpretable feature families such as background and demographics, enrollment and finance, first-semester academic progress, second-semester academic progress, assessment performance, course registration, and virtual learning environment activity

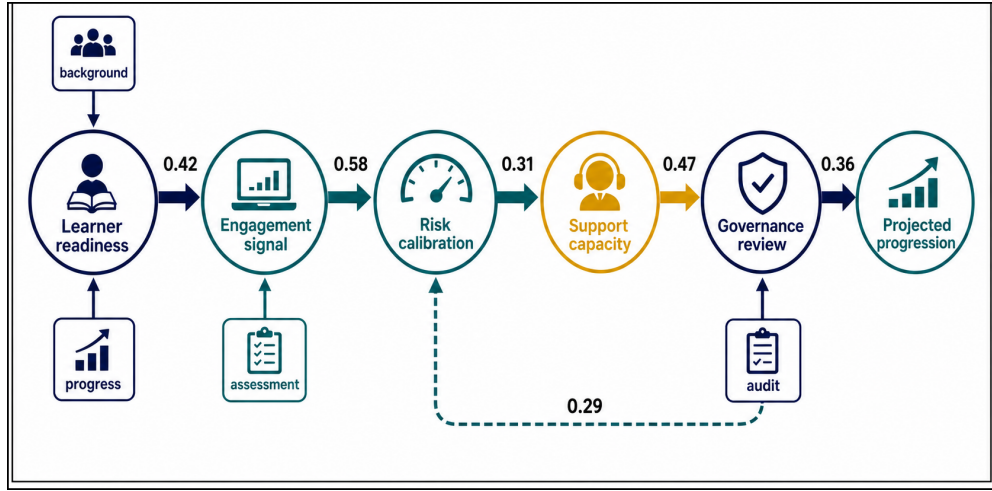


Fig. 3 Analytic path model for capacity-aware adaptive support. Solid arrows show the main evidence flow from readiness and engagement indicators to calibrated risk, support capacity, governance review, and projected progression; the dashed arrow denotes the governance feedback loop to risk calibration

windows. The Open University Learning Analytics Dataset (OULAD) stream includes 25%, 50%, 75%, and 100% course-window engagement summaries, allowing the analysis to examine how early signals change as more of a course unfolds.

Figure 3 expresses the framework as a capacity-aware analytic path. The diagram communicates the order of evidence transformation from learner readiness and engagement signals to calibrated risk, support capacity, governance review, and projected progression. The numeric labels are conceptual markers for the analytic flow and are not interpreted as coefficients, causal effects, or structural path estimates.

The third layer is explainable risk modeling. WALA-CC compares simple baselines with regularized linear and ensemble models, reports discrimination and calibration, and uses feature-level and family-level explanation to describe what the strongest models use. This is intended to support institutional interpretation rather than to automate decisions.

The fourth layer is calibration and scenario analytics. Calibrated probabilities are converted into risk thresholds and support queues. Scenario outputs are reported in operational terms: learners flagged per 100, projected successful progressions per 100 under assumed support intensity, support hours per 100, hours per additional projected success, scenario number-needed-to-support under assumed effect, and subgroup flag-gap diagnostics. The final layer is human capacity and governance. It asks whether the proposed threshold is interpretable, staffable, auditable, non-punitive, and consistent with equity-sensitive review.

4 Data and Methods

4.1 Public evidence streams

The analysis uses four public evidence streams with distinct units of analysis. Table 1 documents the data source, unit, record count, feature count, and analytic role. The learner-level stream uses six open datasets: the Open University Learning Analytics Dataset (OULAD) (Kuzilek, Hlosta, & Zdrahal, 2017), the UCI Predict Students’ Dropout and Academic Success dataset (Realinho, Machado, Baptista, & Martins, 2021), the UCI Student Performance dataset (Cortez, 2008), the UCI Higher Education Students Performance Evaluation dataset (Yilmaz & Sekeroglu, 2022), the UCI Student Academics Performance dataset (Hussain, 2018), and the xAPI-Edu-Data engagement dataset (Abu Amrieh, Hamtini, & Aljarah, 2016). Together, these learner sources provide 38,817 records spanning student-program progression, course outcomes, assessment performance, digital engagement, attendance, family context, and academic-progress variables. OULAD and UCI are treated as the primary public-benchmark evidence because they have the largest learner samples and the richest progress indicators. StudentPerf and xAPI provide secondary external triangulation, while HESPE and SAP are small-sample stress tests. The institution stream uses public College Scorecard indicators for two-year institutions (U.S. Department of Education, 2026). The workforce-context stream uses the NCES CIP-SOC crosswalk and O*NET skill descriptors (National Center for Education Statistics, 2020; National Center for O*NET Development, 2026).

Learner outcomes were recoded into binary successful progression labels. In OULAD, pass and distinction were coded as successful progression, while fail and withdrawn were coded otherwise. In the UCI student-program data, graduate was coded as successful progression in the main specification, while dropout and enrolled were coded otherwise. Because enrolled status is right-censored rather than equivalent to dropout, a sensitivity analysis excluded enrolled cases and compared graduate versus dropout outcomes. In StudentPerf, final course grade at or above 10 was coded as successful progression. In HESPE, grade categories at or above 3 were coded as successful progression. In SAP, Good, Very Good, and Best end-semester performance labels were coded as successful progression. In xAPI-Edu-Data, middle and high performance classes were coded as successful progression. This outcome construction supports comparable risk modeling while preserving the fact that the datasets represent different educational settings.

4.2 Descriptive profile and feature blocks

Table 2 summarizes the learner benchmark profile. The largest sources were OULAD (32,593 records; 15,385 successful progression cases) and UCI (4,424 records; 2,209 successful progression cases). The smaller StudentPerf, HESPE, SAP, and xAPI datasets broaden the sensitivity base by adding secondary-performance, higher-education course, academic-performance, and classroom-engagement evidence. The datasets are intentionally complementary: UCI provides richer enrollment, finance, and semester-progress

Table 1 Expanded public data inventory and analytic role

Data	Source	Unit	Records	Feature count	Analytic role
OULAD	Open University Learning Analytics Dataset	Learner record	32,593	94	Risk modeling, calibration, subgroup diagnostics
UCI	UCI Predict Students' Dropout and Academic Success	Learner record	4,424	36	Risk modeling, calibration, subgroup diagnostics
StudentPerf	UCI Student Performance	Learner record	1,044	33	Risk modeling, calibration, subgroup diagnostics
HESPE	UCI Higher Education Students Performance Evaluation	Learner record	145	31	Risk modeling, calibration, subgroup diagnostics
SAP	UCI Student Academics Performance	Learner record	131	21	Risk modeling, calibration, subgroup diagnostics
xAPI	xAPI-Edu-Data public engagement dataset	Learner record	480	16	Risk modeling, calibration, subgroup diagnostics
Scorecard	U.S. Department of Education	Institution	1,523	5	Two-year context and institutional clusters
CIP-SOC / O*NET	NCES and O*NET	Program, occupation, skill	Public files	–	Workforce-context auditability

indicators; OULAD provides time-window engagement traces; xAPI provides compact digital-engagement measures; and the smaller academic-performance datasets test whether the pipeline remains stable under different variable structures.

Table 2 Learner benchmark descriptive profile

Dataset	N	Success n	Success rate	Predictors	Feature blocks	Subgroup fields
OULAD	32,593	15,385	0.472	94	7	5
UCI	4,424	2,209	0.499	36	4	5
StudentPerf	1,044	814	0.780	33	4	5
HESPE	145	78	0.538	31	4	5
SAP	131	104	0.794	21	4	5
xAPI	480	353	0.735	16	4	5

Table 3 reports the feature-block construction. Numeric variables were median-imputed. Categorical variables were mode-imputed and one-hot encoded. Standardization was used inside models that required scale-sensitive preprocessing. For OULAD, virtual learning environment events were summarized by activity type and by elapsed

course window at 25%, 50%, 75%, and 100% of the course timeline. These early-window features allow the evaluation to distinguish between signals that are available early enough for intervention planning and signals that mainly explain outcomes after most of the course has already passed.

Table 3 Feature-block construction for prediction, ablation, and explanation

Dataset	Feature block	Variables
OULAD	Assessment	8
	Background	6
	Registration	2
	VLE 100%	20
	VLE 25%	19
	VLE 50%	19
	VLE 75%	20
UCI	Background	12
	Enrollment/finance	12
	First-semester	6
	Second-semester	6
StudPerf	Background	16
	Family/socioeconomic	7
	Interim progress	2
	Study/support	8
HESPE	Registration	1
	Family/socioeconomic	6
	Habits/engagement	14
	Access context	10
SAP	Background	5
	Family/socioeconomic	7
	Prior/current progress	4
	School/support	5
xAPI	Background	3
	Course/section	5
	Digital engagement	4
	Support/attendance	4

4.3 Reproducibility and unit separation

The analysis was run through a scripted Python pipeline that writes raw downloads, processed datasets, model outputs, tables, figures, logs, and a numeric claim ledger to a reproducibility folder outside the Overleaf manuscript tree. The manuscript repository contains only LaTeX source, bibliography, figures, and table files needed for submission.

The boundary of inference is explicit. The learner records are public benchmarks from varied educational settings and are not U.S. community-college administrative records. College Scorecard indicators describe U.S. two-year institutional context, and CIP-SOC/O*NET records describe program-occupation-skill context. This role-based

linkage is methodologically conservative: it permits multi-source evidence while avoiding a synthetic learner table that would falsely imply observed workforce or institutional attributes for individual learner records. The resulting evidence is appropriate for pre-pilot pressure testing, not for direct local deployment without institution-specific validation.

Generative AI assistance was used during manuscript drafting, code refinement, figure ideation, and editorial revision under human supervision. All analyses reported here were executed through the reproducible Python pipeline, and all numerical claims were checked against generated output files.

5 Predictive Modeling and Diagnostics

5.1 Model suite

The modeling stream compares two baselines and seven machine-learning estimators. The majority baseline predicts the observed majority class. The stratified warning baseline produces a simple data-informed warning score using the strongest available progress or engagement feature, creating a practical reference point beyond the majority class. The main estimators are elastic-net logistic regression, random forest, extra trees, histogram gradient boosting, calibrated gradient boosting, XGBoost, and LightGBM (Chen & Guestrin, 2016; Ke et al., 2017). This model set was selected to compare transparent regularized linear modeling with tree ensembles commonly used for structured educational records.

5.2 Evaluation design

Each learner dataset was split into stratified 70% training and 30% held-out test partitions with a fixed random seed. All preprocessing steps were fit inside the training pipeline: numeric variables were median-imputed, categorical variables were mode-imputed and one-hot encoded, and scaling was applied only for scale-sensitive estimators. This design prevents held-out test information from entering imputation, encoding, scaling, or model fitting. Training performance stability was evaluated with repeated stratified 5x5 cross-validation. Held-out uncertainty was estimated by 300 bootstrap resamples of the test predictions, using percentile intervals. Receiver operating characteristic curves present area-under-curve evidence visually, while the operating tables report average precision, accuracy, precision, recall, F1, macro-F1, Cohen’s kappa, Brier score, expected calibration error, calibration slope, calibration intercept, and confusion-matrix counts. Expected calibration error used equal-width probability bins. F1, macro-F1, and subgroup diagnostics use the 0.50 probability threshold unless a threshold curve is being analyzed separately. Brier score, expected calibration error, slope, and intercept describe probability quality; kappa provides an agreement-adjusted classification diagnostic.

Paired model-comparison tests were conducted on repeated cross-validation macro-F1 values. For each dataset, the reference model with the highest mean repeated macro-F1 was compared with major comparators using paired t statistics, p values, Cohen’s

d_z , and the number of folds. These tests should be interpreted as repeated-resampling diagnostics rather than as claims that one model is universally superior.

5.3 Ablation, explainability, and subgroup diagnostics

Feature-family ablation estimated how much predictive signal was available from different blocks of variables. Across datasets, the blocks included background and demographics, family and socioeconomic context, enrollment and finance, first- and second-semester progress, assessment performance, study behavior and support, attendance, course registration, and digital engagement. For OULAD, virtual learning environment activity was summarized at 25%, 50%, 75%, and 100% course windows. Explainability was estimated with model-based and permutation-style feature importance, then aggregated into feature-family contributions. This grouped view is more useful for education decision making than a list of individual encoded variables alone (Lundberg & Lee, 2017).

Subgroup diagnostics were computed where dataset variables permitted comparison and subgroup sample sizes were sufficient. The available fields included gender or sex, age band, disability, deprivation-band proxy, debtor and scholarship status, attendance, work status, school context, and absence group, depending on the dataset. For each available subgroup, the pipeline reports observed success, selection rate at the 0.50 risk threshold, precision, recall, false-positive rate, false-negative rate, expected calibration error, calibration-in-the-large, calibration slope, and the maximum selection gap within the variable. False negatives are reported because an early-alert system can fail learners by not offering outreach when support may be useful. These diagnostics are screening evidence for responsible evaluation, not a complete local fairness audit (Barocas, Hardt, & Narayanan, 2023; Holstein, Wortman Vaughan, Daume III, Dudik, & Wallach, 2019).

6 Capacity-Aware Scenario Analysis

The scenario analysis translates calibrated learner probabilities into operational quantities. It is decision analytic: the goal is not to estimate a causal treatment effect, but to show how different alert thresholds and support-intensity assumptions change the size of the support queue, advisor workload, projected progression, and subgroup flag gaps. The UCI learner stream is used for the main scenario analysis because its outcome definition and enrollment-progress variables are closest to the student-program planning problem. Table 4 makes the support assumptions visible before the algorithm is introduced. The low and medium settings represent light-touch and proactive advising/planning scenarios; the high setting is an intensive upper-bound sensitivity case informed by comprehensive student-success programs such as CUNY ASAP rather than by ordinary analytics alerts (Gupta, 2017; Weiss et al., 2019).

The analysis evaluates thresholds from 0.10 to 0.90 and three support-intensity assumptions. The low-intensity assumption applies a small log-odds improvement and 5 support minutes per flagged learner. The medium-intensity assumption applies a moderate log-odds improvement and 15 support minutes. The high-intensity assumption applies a larger log-odds improvement and 30 support minutes. Scenario intervals were

Table 4 Support-intensity assumptions for decision-analytic scenario analysis. Probability changes are absolute changes in projected successful-progression probability at selected baseline probabilities

Intensity	Log-odds delta	Minutes	Change at .20	Change at .50	Change at .80	Evidence anchor
low	0.070	5	0.011	0.017	0.011	Light-touch alert review; smaller than intensive advising evidence
medium	0.150	15	0.025	0.037	0.023	Proactive advising/planning scenario; closer to technology-supported advising
high	0.250	30	0.043	0.062	0.037	Intensive wraparound upper-bound scenario; not a causal effect estimate

Algorithm 1 WALA-CC capacity-aware support scenario

Require: Calibrated success probability p_i , risk threshold τ , assumed log-odds support effect δ , support minutes m

- 1: **for** each learner i **do**
 - 2: Compute risk $r_i = 1 - p_i$
 - 3: **if** $r_i \geq \tau$ **then**
 - 4: Set adjusted probability $p'_i = \text{logit}^{-1}(\text{logit}(p_i) + \delta)$
 - 5: Add $m/60$ support hours
 - 6: **else**
 - 7: Set adjusted probability $p'_i = p_i$
 - 8: **end if**
 - 9: **end for**
 - 10: Report projected success per 100, added success per 100, flagged learners per 100, support hours per 100, hours per additional projected success, scenario number-needed-to-support under assumed effect, and subgroup flag gap
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produced by repeated learner resampling. The outputs are intentionally expressed per 100 learners so that program leaders can compare them with realistic staffing capacity.

7 Results

7.1 Model benchmarking

Table 5 reports held-out operating metrics for the leading benchmark models and baselines. Across the six learner datasets, the leading models reached held-out macro-F1 values from 0.773 in the small HESPE dataset to 0.949 in OULAD. The strongest F1 values were observed for xAPI (0.972), OULAD (0.948), StudentPerf (0.945), SAP (0.938), UCI (0.846), and HESPE (0.800). The table also reports accuracy, precision, recall, Cohen’s kappa, average precision, Brier score, and expected calibration error so that the comparison is not reduced to a single discrimination statistic. Baseline rows are retained because they show how much signal is added by the learner records

beyond majority-class prediction. The small HESPE and SAP results are interpreted as stress tests rather than as evidence of the same strength as OULAD and UCI.

Table 5 Classifier operating metrics for benchmark and leading models. Acc., Prec., Rec., and GB denote accuracy, precision, recall, and gradient boosting

Data	Model	Acc.	Prec.	Rec.	F1	Macro-F1	Kappa	Avg. prec.	Brier	ECE
HESPE	Elastic-net	0.773	0.769	0.833	0.800	0.768	0.538	0.792	0.189	0.149
	Calibrated GB	0.773	0.850	0.708	0.773	0.773	0.549	0.899	0.147	0.080
	XGBoost	0.682	0.708	0.708	0.708	0.679	0.358	0.916	0.166	0.213
	Majority	0.545	0.545	1.000	0.706	0.353	0.000	0.545	0.455	0.455
OULAD	LightGBM	0.949	0.921	0.976	0.948	0.949	0.898	0.982	0.038	0.008
	Hist GB	0.949	0.924	0.972	0.947	0.949	0.898	0.982	0.039	0.005
	XGBoost	0.944	0.916	0.970	0.942	0.944	0.887	0.980	0.042	0.006
	Elastic-net	0.934	0.912	0.951	0.931	0.934	0.867	0.969	0.051	0.021
	Majority	0.528	0.000	0.000	0.000	0.346	0.000	0.472	0.472	0.472
SAP	Extra trees	0.900	0.938	0.938	0.938	0.844	0.688	0.983	0.111	0.194
	Calibrated GB	0.850	0.882	0.938	0.909	0.740	0.483	0.979	0.104	0.165
	RF	0.850	0.906	0.906	0.906	0.766	0.531	0.980	0.121	0.180
	Elastic-net	0.850	0.964	0.844	0.900	0.800	0.605	0.980	0.129	0.173
	Majority	0.800	0.800	1.000	0.889	0.444	0.000	0.800	0.200	0.200
StudPerf	Calibrated GB	0.914	0.947	0.943	0.945	0.875	0.751	0.979	0.063	0.033
	XGBoost	0.908	0.950	0.931	0.940	0.869	0.738	0.979	0.071	0.055
	LightGBM	0.901	0.950	0.922	0.936	0.861	0.722	0.984	0.083	0.078
	Elastic-net	0.876	0.948	0.890	0.918	0.831	0.664	0.986	0.085	0.076
	Majority	0.780	0.780	1.000	0.877	0.438	0.000	0.780	0.220	0.220
UCI	Elastic-net	0.838	0.805	0.891	0.846	0.838	0.676	0.878	0.118	0.031
	Hist GB	0.835	0.812	0.872	0.841	0.835	0.670	0.877	0.123	0.059
	Calibrated GB	0.835	0.812	0.872	0.841	0.835	0.670	0.879	0.117	0.023
	Majority	0.501	0.000	0.000	0.000	0.334	0.000	0.499	0.499	0.499
xAPI	Calibrated GB	0.958	0.946	1.000	0.972	0.943	0.887	0.995	0.046	0.078
	RF	0.958	0.972	0.972	0.972	0.946	0.893	0.996	0.054	0.119
	XGBoost	0.944	0.930	1.000	0.964	0.923	0.847	0.994	0.051	0.057
	Elastic-net	0.931	0.944	0.962	0.953	0.909	0.818	0.992	0.051	0.050
	Majority	0.736	0.736	1.000	0.848	0.424	0.000	0.736	0.264	0.264

Figure 4 presents the receiver operating characteristic (ROC) curves for each learner dataset. AUC values are intentionally shown in the ROC legends rather than as a separate table column. The curves show strong ranking performance in the larger OULAD, UCI, StudentPerf, and xAPI streams, while the smaller HESPE and SAP streams show wider step patterns that are consistent with their limited sample sizes.

Table 6 reports paired repeated-validation comparisons on macro-F1. The reference model in each dataset is the model with the highest mean repeated macro-F1. Differences are positive when the reference model exceeds the comparator. The results show that the strongest models usually improved over elastic-net logistic regression and simpler tree baselines, although some differences were small. For example, LightGBM exceeded

histogram gradient boosting by only 0.001 mean macro-F1 in OULAD, suggesting a robust performance envelope rather than a single model that is universally superior.

Table 6 Paired repeated-validation model comparison on macro-F1

Data	Best model	Comparator	Mean diff.	t	p	Cohen dz	Folds
HESPE	XGBoost	Elastic-net	0.151	6.807	0.000	1.361	25
	XGBoost	RF	0.101	4.969	0.000	0.994	25
	XGBoost	Extra trees	0.108	5.813	0.000	1.163	25
	XGBoost	Hist GB	0.058	3.592	0.001	0.718	25
OULAD	LightGBM	Elastic-net	0.015	26.170	0.000	5.234	25
	LightGBM	RF	0.009	24.519	0.000	4.904	25
	LightGBM	Extra trees	0.011	22.053	0.000	4.411	25
	LightGBM	Hist GB	0.001	2.208	0.037	0.442	25
SAP	RF	Elastic-net	0.090	4.818	0.000	0.964	25
	RF	Extra trees	0.044	2.608	0.015	0.522	25
	RF	Hist GB	0.051	2.088	0.048	0.418	25
StudPerf	RF	Elastic-net	0.037	9.151	0.000	1.830	25
	RF	Extra trees	0.064	9.512	0.000	1.902	25
	RF	Hist GB	0.021	4.847	0.000	0.969	25
UCI	XGBoost	Elastic-net	0.009	3.842	0.001	0.768	25
	XGBoost	RF	0.010	6.260	0.000	1.252	25
	XGBoost	Extra trees	0.010	5.218	0.000	1.044	25
	XGBoost	Hist GB	0.001	1.318	0.200	0.264	25
xAPI	RF	Elastic-net	0.015	2.134	0.043	0.427	25
	RF	Extra trees	0.012	2.018	0.055	0.404	25
	RF	Hist GB	0.026	3.341	0.003	0.668	25

Because the UCI dataset includes an enrolled category that may be right-censored, Online Resource 1 reports an outcome-coding sensitivity analysis. Under the main graduate versus dropout/enrolled coding, the best UCI AUC was 0.910 and the best macro-F1 was 0.838. When enrolled cases were excluded and the task compared graduate versus dropout only, the best AUC was 0.949 and the best macro-F1 was 0.902. The direction of the main conclusion did not reverse: UCI remains predictive, but its reported difficulty depends meaningfully on how ongoing enrollment is treated.

7.2 Feature-family signal and calibration

Table 7 shows that feature families differed sharply in predictive contribution. In UCI, all features produced F1 of 0.838, while second-semester progress alone produced F1 of 0.822 and first-semester progress produced F1 of 0.785. In OULAD, assessment performance alone produced F1 of 0.944, while virtual learning environment activity strengthened across the course timeline from F1 of 0.746 at the 25% window to 0.869 at the 100% window. In StudentPerf, interim academic progress alone slightly exceeded the all-feature row on F1, which reinforces the importance of directly observed academic

progress. These patterns are practically important because early engagement signals are weaker than end-of-course traces but are available early enough to support planning.

Table 7 Feature-block ablation results. Acc. and Avg. prec. denote accuracy and average precision

Data	Feature block	Features	Avg. prec.	Acc.	F1	Macro-F1	Brier	ECE
HESPE	All	31	0.916	0.682	0.708	0.679	0.166	0.213
	Registration	1	0.820	0.727	0.667	0.718	0.155	0.029
	Habits/engagement	14	0.727	0.591	0.609	0.590	0.252	0.229
	Access context	10	0.621	0.591	0.591	0.591	0.324	0.332
	Family/socioeconomic	6	0.490	0.477	0.489	0.477	0.349	0.296
OULAD	All	94	0.982	0.949	0.948	0.949	0.038	0.008
	Assessment	8	0.980	0.946	0.944	0.946	0.041	0.006
	VLE 100%	20	0.919	0.871	0.869	0.871	0.094	0.014
	VLE 75%	20	0.888	0.835	0.833	0.835	0.116	0.015
	VLE 50%	19	0.829	0.787	0.790	0.787	0.145	0.019
	VLE 25%	19	0.767	0.733	0.746	0.733	0.173	0.023
	Registration	2	0.536	0.578	0.599	0.577	0.242	0.029
	Background	6	0.569	0.583	0.569	0.583	0.241	0.029
SAP	All	21	0.983	0.900	0.938	0.844	0.111	0.194
	Prior/current progress	4	0.979	0.825	0.881	0.774	0.119	0.158
	School/support	5	0.938	0.750	0.833	0.667	0.170	0.207
	Family/socioeconomic	7	0.942	0.750	0.821	0.702	0.184	0.230
	Background	5	0.886	0.650	0.767	0.533	0.215	0.259
StudPerf	Interim progress	2	0.987	0.882	0.921	0.845	0.080	0.068
	All	33	0.986	0.876	0.918	0.831	0.085	0.076
	Study/support	8	0.875	0.666	0.766	0.590	0.203	0.213
	Background	16	0.841	0.608	0.722	0.529	0.241	0.254
	Family/socioeconomic	7	0.803	0.522	0.631	0.477	0.255	0.267
UCI	All	36	0.884	0.831	0.838	0.831	0.116	0.024
	Second-semester	6	0.857	0.816	0.822	0.816	0.131	0.038
	First-semester	6	0.831	0.783	0.785	0.783	0.148	0.024
	Enrollment/finance	12	0.766	0.721	0.726	0.721	0.185	0.027
	Background	12	0.621	0.622	0.636	0.621	0.234	0.044
xAPI	All	16	0.996	0.958	0.972	0.946	0.054	0.119
	Digital engagement	4	0.976	0.882	0.921	0.844	0.087	0.063
	Support/attendance	4	0.973	0.868	0.907	0.839	0.101	0.087
	Background	3	0.863	0.694	0.782	0.635	0.209	0.181
	Course/section	5	0.772	0.562	0.667	0.515	0.236	0.198

Figure 5 connects model calibration with threshold behavior. The calibration panel shows that datasets differed in Brier score and expected calibration error even when discrimination was strong. The threshold panels show the operational pattern in UCI: lower thresholds flag more learners and produce more projected additional progression, but also increase advisor-hour demand. The subgroup flag-gap panel indicates that threshold choice can shift differential selection across observed groups, which is why threshold selection should be a governance decision rather than a purely technical default.

Figure 6 reports grouped explainability and ablation patterns. In OULAD, assessment performance and later virtual learning environment activity carried the strongest signal. In UCI, academic progress and enrollment/finance variables were prominent. The grouped explanation is useful for institutional planning because it distinguishes support-relevant signals from background variables that may be less appropriate for direct action.

7.3 Subgroup diagnostics and learner profiles

Table 8 reports subgroup diagnostics for available variables, showing the largest selection-gap examples within each learner dataset. The diagnostics are strongest for identifying review needs rather than for making deployment claims. The table adds precision, recall, false-positive rate, and false-negative rate because early-alert systems can harm learners both by over-selecting groups for unnecessary intervention and by missing learners who could benefit from support. In OULAD, deprivation-band groups showed meaningful selection-rate differences at the 0.50 threshold, even when expected calibration error remained low. In xAPI, absence groups showed a larger selection gap, reflecting the close relationship between observed participation and performance classification. These patterns support treating subgroup diagnostics as a routine part of model reporting.

Table 8 Subgroup classification, selection, and calibration diagnostics

Data	Variable	Group	N	Obs.	Sel.	Prec.	Rec.	FPR	FNR	ECE	Sel. gap
HESPE	sex	1	58	0.379	0.362	0.857	0.818	0.083	0.182	0.090	0.293
	sex	2	87	0.644	0.655	0.930	0.946	0.129	0.054	0.062	0.293
	work	1	49	0.469	0.408	1.000	0.870	0.000	0.130	0.084	0.196
OULAD	imd_band	0-10%	3,311	0.352	0.375	0.925	0.987	0.043	0.013	0.026	0.291
	imd_band	10-20	3,516	0.386	0.409	0.929	0.983	0.047	0.017	0.019	0.291
	imd_band	20-30%	3,654	0.407	0.430	0.938	0.989	0.045	0.011	0.021	0.291
SAP	attendance	Average	47	0.830	0.830	0.974	0.974	0.125	0.026	0.167	0.357
	attendance	Good	56	0.893	0.893	1.000	1.000	0.000	0.000	0.129	0.357
	attendance	Poor	28	0.536	0.536	0.933	0.933	0.077	0.067	0.178	0.357
StudPerf	age_band	16-17	558	0.805	0.765	0.979	0.931	0.083	0.069	0.053	0.154
	age_band	18-19	278	0.698	0.640	0.966	0.887	0.071	0.113	0.061	0.154
	age_band	<=15	194	0.830	0.794	0.981	0.938	0.091	0.062	0.054	0.154
UCI	scholarship	0	3,325	0.413	0.438	0.831	0.881	0.127	0.119	0.030	0.381
	scholarship	1	1,099	0.760	0.820	0.893	0.964	0.364	0.036	0.027	0.381
	debtor	0	3,921	0.538	0.576	0.856	0.917	0.180	0.083	0.028	0.380
xAPI	absence	Above-7	191	0.393	0.330	0.952	0.800	0.026	0.200	0.130	0.629
	absence	Under-7	289	0.962	0.958	0.989	0.986	0.273	0.014	0.085	0.629
	gender	F	175	0.863	0.846	0.993	0.974	0.042	0.026	0.087	0.216

Learner-profile analysis adds another layer of interpretation. Table 9 reports the selected profile solution. OULAD separated into a low-risk high-engagement profile (16.1% of learners; observed success 0.868; 2.73 support hours per 100 at the medium threshold) and a larger higher-risk profile (83.9%; observed success 0.396; 14.56 support hours per 100). UCI separated into four profiles, including a high-risk profile with observed success 0.100 and 23.52 medium-intensity support hours per 100. HESPE, SAP, StudentPerf, and xAPI also produced distinct high- and low-risk profiles despite

their smaller sample sizes. Figure 7 shows the corresponding profile signatures and predicted-risk distribution.

Table 9 Learner profile solution and support implications

Data	Profile	N	Share	Success	Risk	Flagged	Hours/100	Dominant signature
HESPE	1	30	0.207	0.200	0.818	0.867	21.667	Course registration
HESPE	2	77	0.531	0.883	0.115	0.052	1.299	Predicted risk
HESPE	3	38	0.262	0.105	0.886	0.974	24.342	Predicted risk
OULAD	1	5,254	0.161	0.868	0.129	0.109	2.726	VLE activity
OULAD	2	27,339	0.839	0.396	0.600	0.582	14.558	100% Predicted risk
SAP	1	79	0.603	1.000	0.110	0.000	0.000	Predicted risk
SAP	2	25	0.191	0.080	0.832	1.000	25.000	Predicted risk
SAP	3	27	0.206	0.852	0.361	0.074	1.852	Predicted risk
StudentPerf		720	0.690	0.983	0.037	0.000	0.000	Interim academic progress
StudentPerf		324	0.310	0.327	0.804	0.852	21.296	Predicted risk
UCI	1	603	0.136	0.360	0.640	0.642	16.045	Predicted risk
UCI	2	2,122	0.480	0.796	0.208	0.113	2.828	Predicted risk
UCI	3	1,440	0.325	0.100	0.889	0.941	23.524	Predicted risk
UCI	4	259	0.059	0.614	0.369	0.320	8.012	First-semester progress
xAPI	1	180	0.375	0.306	0.696	0.778	19.444	Predicted risk
xAPI	2	300	0.625	0.993	0.084	0.000	0.000	Digital engagement

7.4 Institutional context, workforce-context auditability, and capacity trade-offs

Table 10 summarizes the contextual evidence layer. The College Scorecard stream produced three two-year institution clusters with distinct enrollment and completion profiles. The workforce stream showed that complex problem solving, process skills, and

basic content skills had the highest mean skill-importance values among the analyzed O*NET families. These indicators do not measure individual learner employment. Their role is to make the workforce-readiness rationale auditable and to prevent the pathway framing from resting only on predictive accuracy.

Table 10 Institutional and workforce-context audibility

Context	Group	N	Enroll.	Comp./imp.	Earnings/source
Institution cluster	Cluster 1	135	13,625	0.160	32,100
	Cluster 2	564	2,506	0.206	28,500
	Cluster 3	173	345	0.600	29,700
Skill family	Complex problem solving	408		3.459	O*NET IM scale
	Process skills	1,632		3.394	O*NET IM scale
	Basic content skills	2,448		3.347	O*NET IM scale
	Systems skills	1,224		3.139	O*NET IM scale
	Social skills	2,448		3.092	O*NET IM scale

Table 11 reports the medium-intensity scenario sensitivity results. At the 0.20 threshold, the model flagged 62.9 learners per 100 and required 15.73 support hours per 100, with 1.10 projected additional successful progressions per 100. At the 0.50 threshold, it flagged 46.7 learners per 100, required 11.67 support hours per 100, and projected 0.60 additional successful progressions per 100. At the 0.80 threshold, it flagged 34.8 learners per 100, required 8.70 support hours per 100, and projected 0.20 additional successful progressions per 100. The hours-per-additional-success metric rose as thresholds became more restrictive, illustrating the cost of narrower support coverage.

Table 11 Capacity-aware medium-intensity scenario sensitivity. Success, added, flagged, and hours are per 100 learners

Threshold	Success	Added	Flagged	Hours	Hours/add	Scenario NNS	Flag gap
0.200	51.290	1.100	62.930	15.732	14.315	57.261	0.217
0.300	51.059	0.912	55.823	13.956	15.309	61.234	0.238
0.500	50.782	0.596	46.675	11.669	19.612	78.449	0.239
0.700	50.448	0.299	38.444	9.611	32.176	128.7	0.220
0.800	50.400	0.195	34.818	8.705	44.622	178.5	0.220

Figure 8 visualizes the capacity frontier across thresholds and support-intensity assumptions. The key result is not that one threshold is universally optimal. Rather, the frontier makes the design decision explicit: lower thresholds offer broader coverage

and larger projected gains under the assumed effect, whereas higher thresholds reduce the support queue but produce fewer projected additional progressions.

Figure 9 complements the capacity analysis by showing the workforce-skill layer used to contextualize pathway readiness. As a worked example, a college considering an information-technology support pathway could use CIP-SOC links and O*NET skill-family evidence to document that complex problem solving, process skills, and basic content skills are salient in the occupational context, then ask whether early support thresholds create enough advising capacity for learners in the relevant program. This use is deliberately contextual: it does not infer an individual learner’s job placement, wage, or workforce readiness.

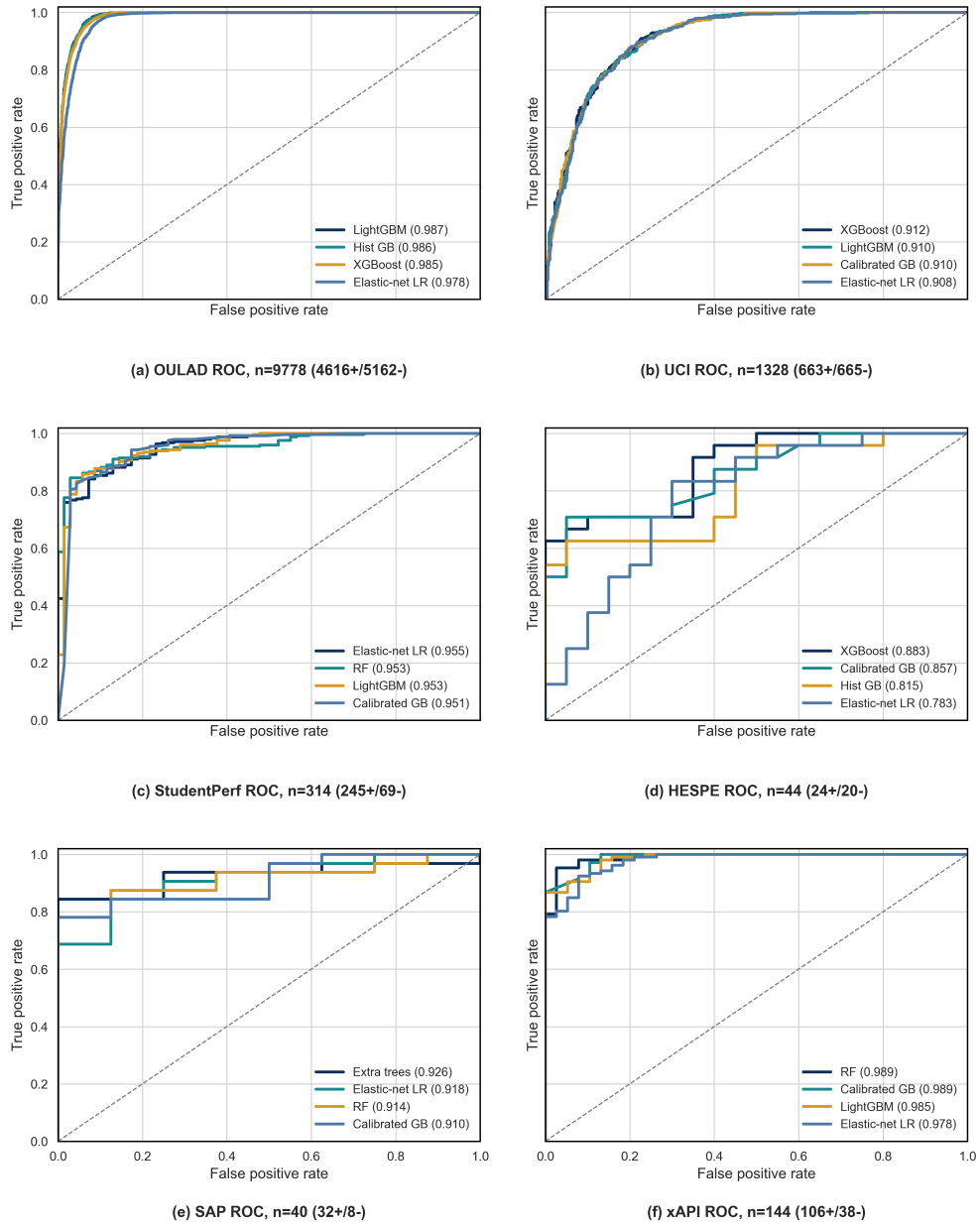


Fig. 4 ROC curves for the six learner datasets. FPR denotes false positive rate, TPR denotes true positive rate, and the value in parentheses in each legend entry is the held-out AUC for that model. Panel labels report held-out sample size with positive/negative outcome counts; the more visibly stepped HESPE and SAP curves reflect their smaller held-out samples

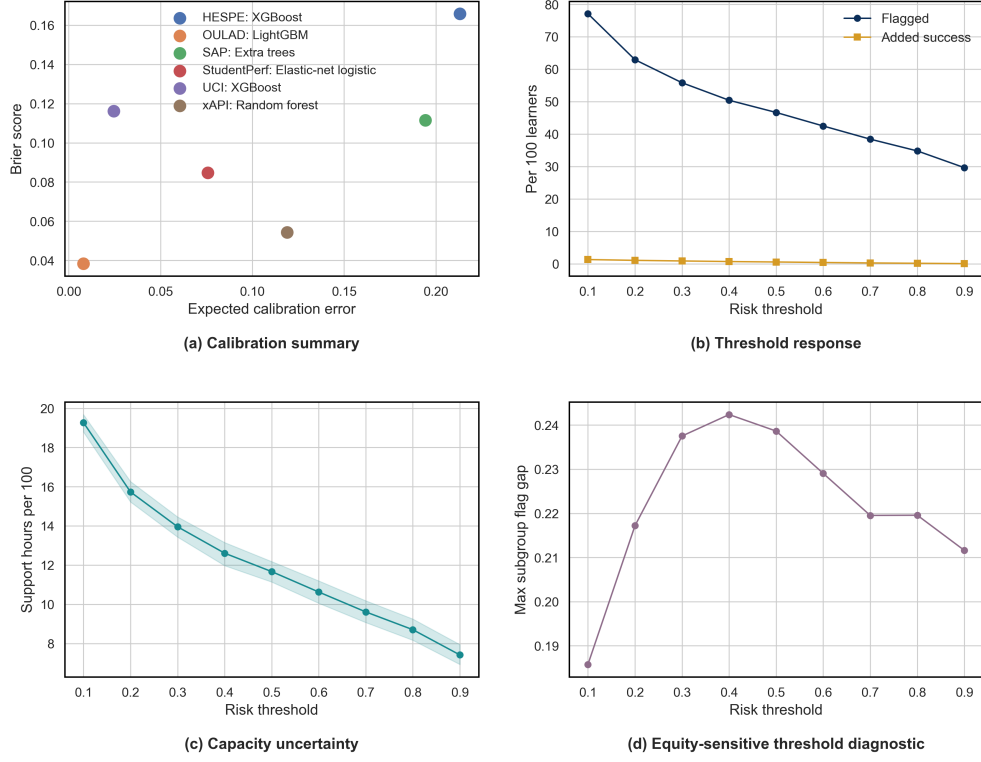
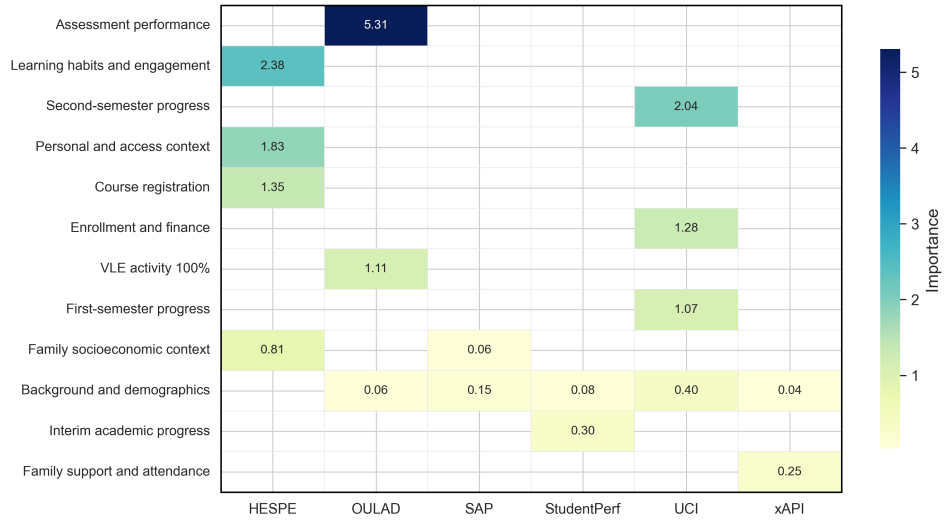
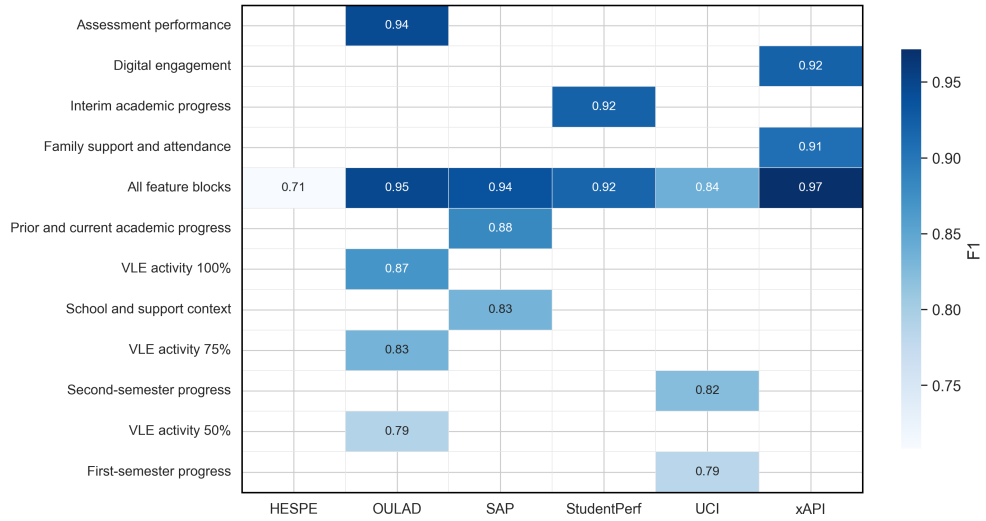


Fig. 5 Calibration and threshold diagnostics. ECE denotes expected calibration error; Brier denotes Brier score; panels show calibration quality, threshold response, advisor-hour uncertainty, and the maximum subgroup flag-gap diagnostic across risk thresholds

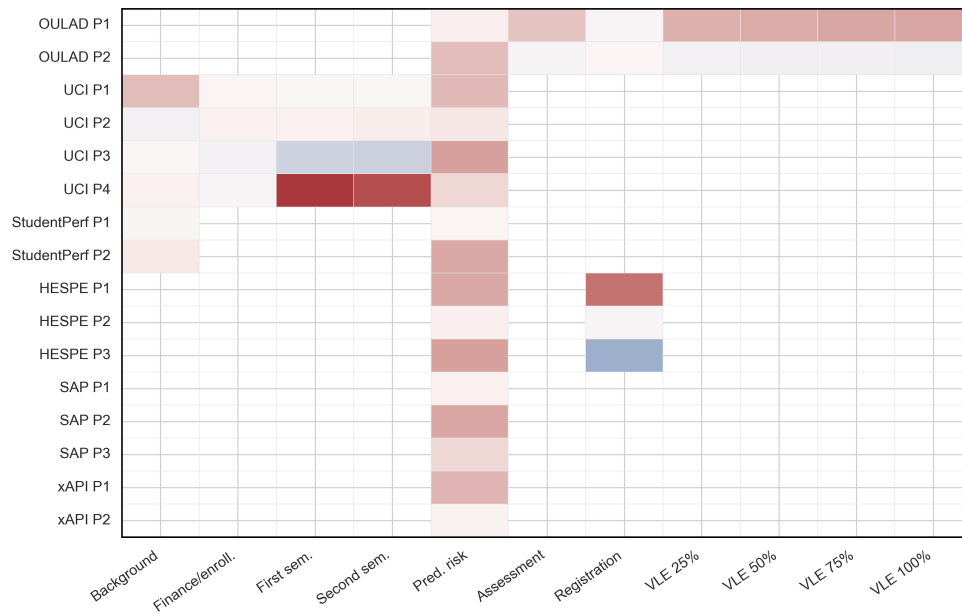


(a) Grouped explainability signal

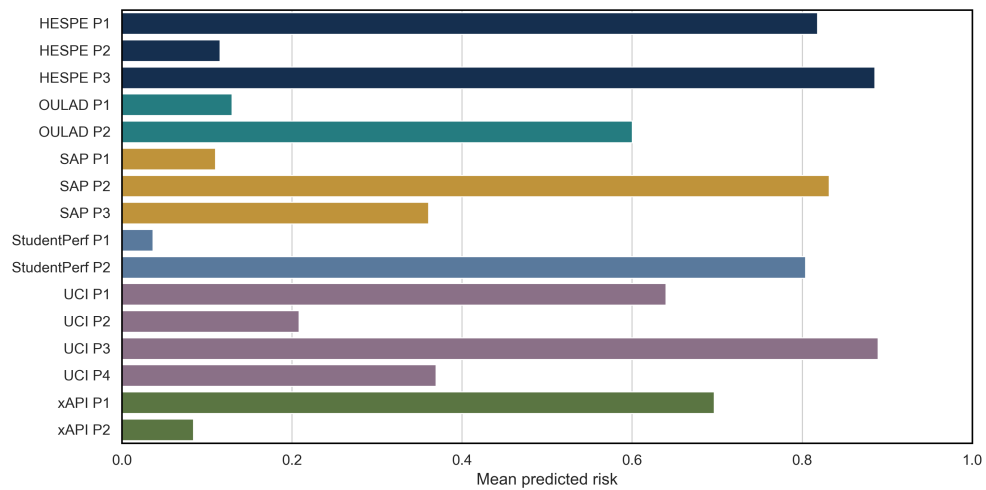


(b) Feature-block ablation by F1

Fig. 6 Explainability and feature-block ablation dashboard. Panel (a) reports grouped feature-family importance from the strongest learner models; panel (b) reports held-out F1 for the strongest feature blocks within each dataset

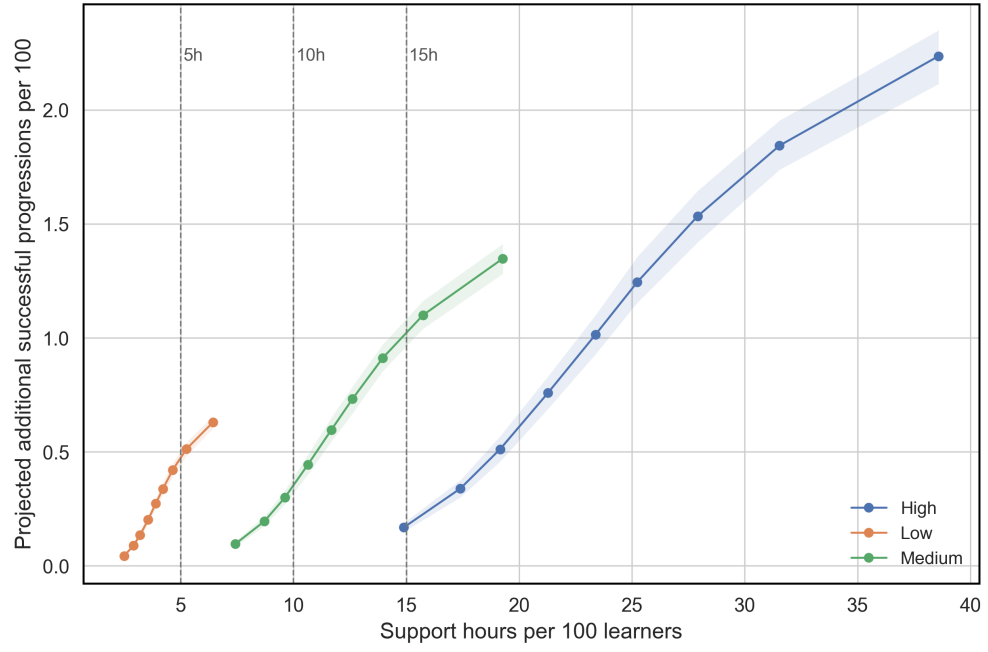


(a) Learner profile signatures



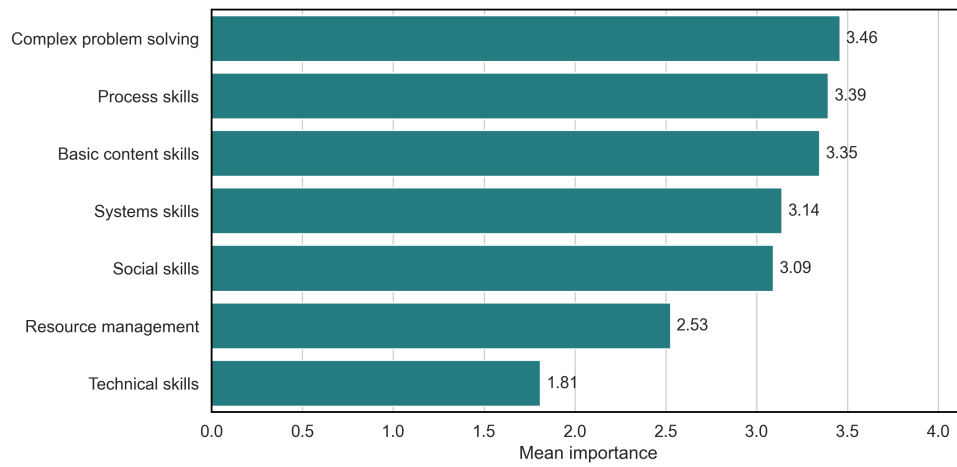
(b) Predicted risk by profile

Fig. 7 Learner profile signatures and predicted risk. Panel (a) shows standardized profile signatures, where warmer values indicate higher profile means and cooler values indicate lower profile means; panel (b) shows mean predicted risk by profile

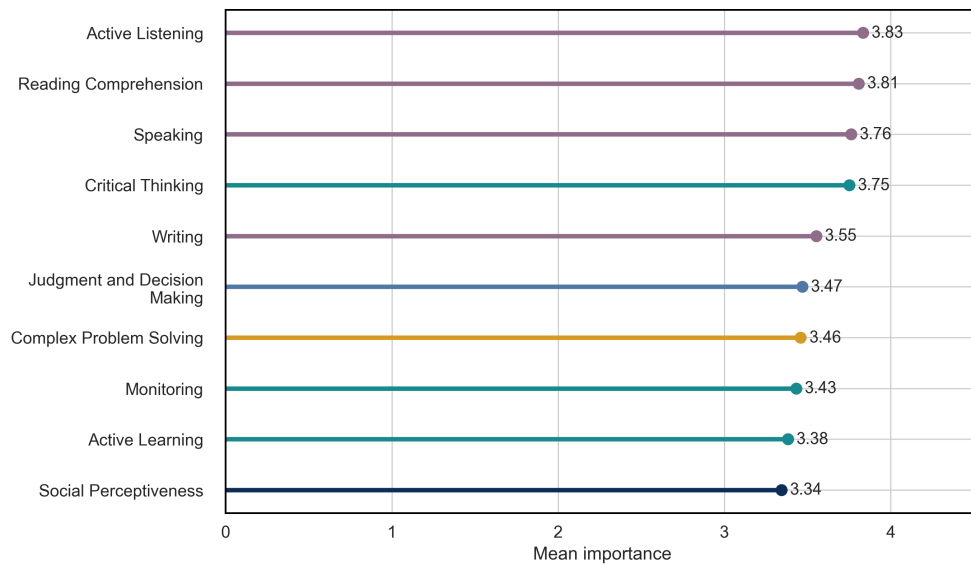


(a) Capacity-aware Pareto frontier

Fig. 8 Capacity-aware scenario frontier. The panel shows projected additional successful progressions and advisor-hour requirements across thresholds and support-intensity assumptions; vertical dashed lines mark example advisor-hour budgets per 100 learners



(a) O*NET skill-family intensity



(b) Top O*NET skill links

Fig. 9 Workforce-context auditability. Panel (a) reports mean O*NET skill-family importance; panel (b) reports the highest-importance skill links used as contextual evidence for workforce-oriented pathway scenarios

8 Discussion

8.1 From prediction to pilot readiness

The results show that open learner benchmark records can support technically strong progression-risk modeling, but the more important finding is that model performance alone is not enough to justify learner-facing use. OULAD and UCI provide the strongest public-benchmark evidence because of their sample size and progress indicators; StudentPerf and xAPI provide useful triangulation; HESPE and SAP should be read as small-sample stress tests. Across these sources, the most institutionally useful evidence combines discrimination, calibration, early-action timing, subgroup diagnostics, and support capacity.

8.2 Early signals and action timing

The ablation results show why early-action validity must be separated from retrospective predictability. In OULAD, assessment performance and later virtual learning environment windows were highly predictive, but the 25% and 50% engagement windows are more relevant for planning timely support. In UCI, first-semester progress carried meaningful signal before second-semester variables became available. The implication is not that institutions should wait for the strongest late predictor. Rather, they should inspect whether earlier, less complete signals are good enough for low-stakes supportive inquiry.

8.3 Explanation, subgroup diagnostics, and learner agency

Feature-family explanations help translate models into educational decisions without turning predictive associations into causal claims. Academic progress, engagement windows, finance variables, and attendance can suggest what kind of conversation might be useful, but they should not be treated as deterministic reasons for failure. Subgroup diagnostics serve a similar screening role. Selection gaps, false-positive rates, false-negative rates, and calibration differences should trigger local review rather than automatic model rejection or approval.

Learner agency should remain visible in any local pilot. Risk scores should be prompts for supportive inquiry, not labels attached to learners. They should not be used to deny access, restrict course choices, punish learners, or reduce opportunities. If analytics influence outreach, learners should receive understandable explanations, opportunities to correct data, ways to contest interpretations, and escalation paths when an alert feels inappropriate. Advisors should combine model outputs with learner voice and contextual knowledge.

8.4 Capacity-aware thresholding

The scenario analysis reframes thresholds as resource-allocation choices. Lower thresholds increase coverage and projected additional progression under the assumed support effect, but they also create larger advising queues. Higher thresholds are easier to staff but leave more learners outside the support boundary. Because the log-odds support

deltas are assumptions, not estimated causal effects, the scenario tables are best read as capacity-planning tools. They help an institution ask what support promise it is making when it chooses a threshold.

8.5 Workforce-context auditability

The workforce-context layer is intentionally modest. CIP-SOC and O*NET evidence can help a pathway director document which program-occupation and skill-family assumptions surround an analytics pilot. They cannot validate learner employment outcomes, wages, licensing, or individual readiness. This distinction preserves the pathway motivation while avoiding unsupported labor-market claims. Local pilots would need institution-specific program maps, employer evidence, credential data, and labor-market information if workforce outcomes became part of the evaluation.

8.6 Challenges to the framework

Several objections are appropriate. First, the learner datasets are not U.S. community-college administrative records. WALA-CC therefore supports public-benchmark pre-pilot evaluation, not local validation. Second, UCI is not a U.S. community-college dataset, and its enrolled label raises a right-censoring concern. The sensitivity analysis shows that conclusions are stronger when enrolled cases are excluded, but local data would still be required. Third, workforce indicators are not learner employment outcomes; they are context for pathway governance. Fourth, scenario support effects are assumed rather than observed. Finally, a local pilot is still necessary. The value of WALA-CC is to help institutions decide whether such a pilot is warranted and what should be monitored during it.

9 Practical Implications

For institutional researchers, the study provides a reporting template for learner-risk analytics. Before recommending a pilot, analysts should report ROC evidence, calibration, expected calibration error, subgroup errors, threshold curves, early-window performance, and a numeric claim ledger that links manuscript claims to reproducible outputs.

For advising leaders, the key implication is that learner-risk analytics should be evaluated as a support-capacity problem. Competing thresholds should be compared against actual advisor hours, outreach workflows, and program priorities. A threshold that looks technically attractive is not ready for use if the institution cannot provide the support it implies.

For pathway directors, feature-family evidence can guide discussion about what kind of help should follow an alert. Early engagement, academic-progress, finance, and attendance signals suggest different support conversations. The model should not decide the intervention; it should help organize a timely, human review.

For governance committees, the framework clarifies what must be documented before learner-facing use: intended users, permissible decisions, learner communication, contestability, data correction, subgroup monitoring, retention rules, and escalation

paths. These requirements are part of technical readiness, not a separate administrative afterthought.

For policy-facing administrators, the findings suggest that analytics procurement should be treated as an implementation and capacity decision rather than a software purchase. The central question is whether an institution can validate, explain, staff, govern, and continuously monitor the system in its local context.

10 Limitations and Future Research

This study has limitations. First, the learner datasets are open benchmark records rather than local administrative records from a specific U.S. community-college system. They are appropriate for reproducible pre-pilot evaluation, but they cannot represent every local program structure, advising practice, financial-aid rule, scheduling constraint, or regional labor-market condition. Second, outcome labels differ across datasets. The UCI enrolled category is especially important because ongoing enrollment is right-censored; the sensitivity analysis reduces this concern but does not eliminate the need for local outcome definitions.

Third, some datasets are small and context-specific, so HESPE and SAP results should be interpreted as stress tests. Fourth, late features may inflate retrospective predictive performance. OULAD assessment variables and later engagement windows are useful for understanding signal growth, but early-window evidence is more relevant for support planning. Fifth, scenario support effects are assumed, not causal. Randomized, quasi-experimental, or carefully staged pilot designs would be needed to estimate the real effect of advising, tutoring, or pathway support.

Sixth, the workforce-context layer uses program-occupation and occupational-skill descriptors rather than observed employment outcomes. It supports auditability, not individual job prediction. Seventh, subgroup diagnostics are screening tools constrained by public variables and do not constitute a complete fairness audit. Local deployment would require richer protected-class review, stakeholder interpretation, monitoring for intersectional harms, and learner communication procedures.

Future research should test the framework with community-college partners, compare staffable alert thresholds, study advisor workflow and trust, evaluate learner-facing explanations and contestability, and connect local pathway analytics to credential, transfer, licensing, and employment outcomes under appropriate privacy protections. A two-site pilot with longitudinal calibration monitoring would be a useful next step.

11 Conclusion

This article presented WALA-CC, a pre-pilot evaluation framework for learner-risk analytics in workforce-oriented postsecondary pathways. The study combined six open learner benchmarks, public two-year institutional indicators, and occupational skill evidence in a role-separated computational evaluation. The empirical pipeline showed strong learner-risk prediction, meaningful calibration differences, early-action trade-offs, feature-family patterns, subgroup diagnostic needs, learner-profile variation, and clear capacity consequences across support thresholds.

The central conclusion is that learner-risk prediction is only one part of institutional readiness. Before piloting analytics for adaptive support, colleges should inspect calibration, action timing, explanation, subgroup behavior, staffing capacity, learner agency, and workforce-context auditability. Public benchmarks can pressure-test the workflow and expose key design questions, but they do not replace local validation. WALA-CC helps institutions ask better deployment questions before moving from analytics design to learner-facing pilots.

Statements and Declarations

Funding

No funding was received.

Competing interests

The authors declare no competing interests.

Ethics approval

This study used publicly available, de-identified learner benchmark, institutional, program, and occupational datasets and did not involve direct interaction with human participants.

Consent to participate

Not applicable.

Consent for publication

Not applicable.

Data availability

The analysis uses public datasets from the UCI Machine Learning Repository, the Open University Learning Analytics Dataset, the U.S. College Scorecard, the NCES CIP-SOC crosswalk, and O*NET. The processed research materials, analysis scripts, search-query and source-access documentation, aggregate results, generated tables and figures, codebook, checksums, excluded-data notes, and supplementary replication artifacts are available in an external repository. Repository details have been provided separately through the journal submission system to preserve double-anonymized review.

Code availability

The analysis code is included in the same external repository described in the data availability statement, with repository details supplied separately through the journal submission system for double-anonymized review.

Use of generative AI

Generative AI assistance was used to support drafting, code generation, figure ideation, and editorial revision under human supervision. The authors reviewed and remain responsible for the manuscript content, analysis, figures, and interpretations.

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